Related Work: GNN-based EMG Gesture Recognition (Group 05, NinaPro DB2)

Title	Authors	Year	Dataset	Application area	Method / Model	Metrics	Strengths	Limitations	Research gap	Relation to our work
Cross-User Electromyography Pattern Recognition Based on a Novel Spatial-Temporal Graph Convolutional Network (IEEE TNSRE)	M. Xu, X. Chen, Y. Ruan, X. Zhang	2024	Own HD-sEMG database: 128 channels, 11 subjects, 17 gestures, 28,424 samples	Myoelectric control / prosthesis; cross-user gesture recognition	CNN-MSTGCN: CNN feature extractor + multi-view (distance graph and adaptive functional graph) spatial-temporal GCN (Chebyshev) with spatial and temporal attention; transfer learning (TL-CMSTGCN)	Recognition rate. Within-subject 96.9%; user-independent 68.0% (ResNet50 47.5%, LSTM 57.1%); with TL 92.3%; inference 2.61 ms/sample	Learns functional links between non-adjacent muscles; ablation shows each module helps; beats ResNet50/LSTM; transfer learning cuts user training burden	User-independent accuracy (68%) still low; offline only; private dataset; needs fine-tuning data for 92.3%	Cross-user generalisation without per-user calibration is unsolved; graph choice not compared systematically	Shows GNNs can beat CNN/LSTM across users. We will test graph variants on a public dataset (NinaPro DB2) with a subject-independent split
Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification (IEEE TNSRE)	H. Lee, M. Jiang, J. Yang, Z. Yang, Q. Zhao	2025	5 public datasets: NinaPro DB5, CapgMyo A and B, UCI EMG, BandMyo	Prosthetic control; hand gesture classification; explainable AI; sensor reduction	GCN (message passing) on explicit graphs built from sensor layout: KNN-1, KNN-2, fully connected, each with or without sensor-wise temporal (SW) edges; also GAT, STGCN, GraphLSTM, GCLSTM	Accuracy, intra-subject and inter-subject. Intra: over 99.3% on all datasets with FC+SW or KNN-2+SW; inter-subject up to 99.4% (BandMyo), 92.4% (DB5), 93.7% (CapgMyo B); sensor-importance maps	Many datasets; compares six graph structures; interpretable sensor importance; stays accurate with 28 of 128 sensors	Random 8:2 split over windows (possible leakage between near-identical windows); dense graphs are costly; no real-time test; NinaPro DB2 not used	Graph structure is fixed by layout; no study on sparse 12-channel setups such as DB2; evaluation protocol may be optimistic	Closest template for our graph-construction comparison (KNN vs fully connected vs temporal edges). We apply it to DB2 with a repetition-wise split
A Novel Multi-Feature Fusion Network With Spatial	F. Hu, M. Qian,	2024	Own MyoData (17	Armband gesture recognition for	MFN-SPSCA: Pearson-correlation graph with 4 partitioning	Accuracy, precision, recall, F1, Kappa, AUC, inference time.	Fuses time, space and frequency features; best	Subject-dependent only; authors state it generalises	No cross-subject evaluation; complexity	Gives a correlation-based graph

Partitioning Strategy and Cross-Attention for Armband-Based Gesture Recognition (IEEE TNSRE)	K. He, W.-A. Zhang, X. Yang		subjects, 11 gestures, Myo armband 8 sEMG + 3 ACC) and NinaPro DB5 (10 subjects, 53 gestures, 2 Myo)	prosthetics and exoskeletons	strategies in a 9-layer STGCN, plus multi-scale CNN-BiLSTM, CWT branch and cross-attention fusion	98.78% (MyoData), 91.60% (DB5); 12.18 ms and 21.25 ms per sample	DB5 results among compared methods; ablations and t-SNE/CAM visualisation; fast enough for real time	poorly to unseen users; large multi-branch model; sEMG+ACC needs IMU	makes it hard to tell what the graph itself contributes	option and DB5 baselines. We keep the model simple to isolate the effect of the graph on raw sEMG
Hand Gesture Recognition From Surface Electromyography Signals With Graph Convolutional Network and Attention Mechanisms (IEEE Sensors J.)	H. Zhou, H. T. Le, S. Zhang, S. L. Phung, G. Alici	2025	NinaPro DB2 (40 intact subjects, 12 Delsys sensors, Exercises B/C/D, 49 gestures) and own UOW dataset (10 subjects, 8-ch armband, 9 gestures, 3 arm positions)	Prosthetic hand control; robustness to limb position	CovGCN: shared prior ring graph refined dynamically by a covariance-based topology refinement module (CovTRM); graph-conv blocks with multi-scale temporal convolution (best at 7 layers)	Classification accuracy. DB2 Ex. B 88.02%, C 81.30%, D 93.30%; UOW 84.69% vs CNN 81.98%; 1.07M parameters, 29.6 ms/signal	Uses NinaPro DB2 raw sEMG directly; dynamic graph; beats CNNs and classical ML; light model; limb-position test	Intact subjects only; offline; subject-specific, no cross-subject test; slightly below a Transformer on Ex. B (88.91%)	First dynamic-graph model on sEMG, but no unseen-subject evaluation and no comparison of alternative graph definitions	Same dataset and window setup (200 ms, 12 channels) as ours; its DB2 numbers are our main benchmark. We add the cross-subject test it leaves open

Graph Neural Networks for HD EMG-based Movement Intention Recognition: An Initial Investigation (IEEE RASSE)	S. M. Massa, D. Riboni, K. Nazarpou	2022	Public HD-sEMG database: 2 x 64 channels, 20 subjects, 65 isometric gestures, 2048 Hz	Amputee movement intention recognition; robotic prosthesis control	Graph with 128 nodes (channels) and 884 distance-based edges; node feature = 32 ms window (65 samples); GraphSAGE (6 layers) + mean pooling + MLP	Classification error rate. Mean 8.75% (std 4.9); 45 of 65 gestures under 10% error	First GNN on HD-EMG; very short 32 ms window (low latency); many gestures	No comparison with other ML/DL methods; per-subject training only; weak on combined multi-DoF gestures; simple heuristic graph	No baselines, no cross-subject test, graph chosen heuristically	Motivates a GNN on raw sEMG windows. We add baselines (CNN/LSTM) and a systematic graph comparison
A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals (arXiv:2607.07850, preprint)	P. Vipulanandan, K. Prematne, M. Murthi	2026	Public Myo-band dataset (from De Silva et al.): 8 subjects, 5 gestures, 8 channels, 200 Hz, 20 repetitions per gesture	Real-time gesture detection for prostheses and AR	Graph per 0.4 s window: nodes = channels, edges = Pearson correlation (fully connected, weighted); event-detection threshold to find gesture onset; simple GNN (2 fully connected layers, ReLU)	Accuracy, confusion matrix, ROC/AUC. Average 98.19% (90% overlap), 99.89% (97.5% overlap); 48 ms per graph+prediction on an M1 Pro CPU	Very fast; includes onset detection; correlation graph needs no layout knowledge; robust to less training data (AUC 0.967 at 70% overlap)	Only 5 gestures and 8 subjects; random split over 90-97.5% overlapping windows (likely leakage, my reading); subject-specific; very simple GNN; preprint, not peer reviewed	Easy task and optimistic protocol; no cross-subject test; no comparison with strong deep baselines	Supports the correlation-graph option and real-time angle. We use a harder task (DB2, 49 gestures) and a leakage-free spli